

Weijie Lyu

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Summary

Forth-year Ph.D. candidate at UC Merced working on computer vision, with current research focus on **video generation** and **3D/4D reconstruction**. Published first-author papers in top-tier CV venues including CVPR, ICCV, ECCV, NeurIPS etc. Open-sourced GitHub projects with 500+ (FaceLift) and 400+ (Gaga) stars. Industry experience at Apple, Adobe Research and ByteDance.

Education

University of California, Merced, *Ph.D. in EECS* *Aug. 2023 – Present*
Advisor: **Prof. Ming-Hsuan Yang**

University of Illinois, Urbana-Champaign, *M.S. in Computer Science* *Aug. 2021 – May 2023*
Advisor: **Prof. Derek Hoiem**
Thesis: *MACON: Memory-Augmented Continual Learning for Open-world Classification*.

ShanghaiTech University, *B.E. in Computer Science and Technology* *Sep. 2017 – Jul. 2021*
Senior Thesis Advisor: **Prof. Zhihao Jiang**
Thesis: *Intelligent Driving Verification Platform Based on Virtual Traffic Conditions*.
Award: Outstanding Graduate of Shanghai, Outstanding Graduate of ShanghaiTech University

University of California, Berkeley, *Visiting Student in EECS* *Aug. 2019 – May 2020*
Undergraduate Research Advisor: **Prof. Sonia Bishop**.

Research Experience

Apple, Cupertino, CA *May 2026 – Aug. 2026*
- **Research Scientist Intern**. Advisor: **Dr. Lu Jiang** and **Dr. Jianyi Wang**.
- Worked agentic auto-research for video generation.

ByteDance, San Jose, CA *Feb. 2026 – May 2026*
- **Research Scientist Intern**. Advisor: **Dr. Yujing Wang** and **Dr. Xiangtai Li**.
- Developed VGC, a video-to-video generation model for geometry correction ([1]).

Adobe Research, San Jose, CA *May 2024 – Jan. 2026*
- **Research Scientist/Engineer Intern**. Advisor: **Dr. Zhixin Shu**.
- Developed FaceCam, a portrait video camera control framework without scale ambiguity ([2]), and FaceLift, a framework for high-fidelity 3D head reconstruction ([5]).

Publications

* denotes equal contribution. Full publication list: **Personal Page** / **Google Scholar**.

- [1] **Weijie Lyu**, Tiancheng Shen, Xiangtai Li, Yujing Wang, Xiaoyuan Wang, Haobo Yuan, Yizhou Zhao, Ming-Hsuan Yang. **Learning to Correct Geometry in Generated Videos**. In *NeurIPS*, 2026.
- [2] **Weijie Lyu**, Ming-Hsuan Yang, and Zhixin Shu. **FaceCam: Portrait Video Camera Control via Scale-Aware Conditioning**. In *CVPR*, 2026. [Project Page](#).
- [3] **Weijie Lyu**, Xueting Li, Abhijit Kundu, Yi-Hsuan Tsai, and Ming-Hsuan Yang. **Gaga: Group Any Gaussians via 3D-aware Memory Bank**. In *TMLR*, 2026. [Project Page](#).
- [4] Jiageng Liu*, **Weijie Lyu***, Xueting Li, Yejie Guo, and Ming-Hsuan Yang. **Edit3r: Instant 3D Scene Editing from Sparse Unposed Images**. In *ECCV*, 2026. [Project Page](#).
- [5] **Weijie Lyu**, Yi Zhou, Ming-Hsuan Yang, and Zhixin Shu. **FaceLift: Learning Generalizable Single Image 3D Face Reconstruction from Synthetic Heads**. In *ICCV*, 2025. [Project Page](#).
- [6] Junqi You, Chieh Hubert Lin, **Weijie Lyu**, Zhengbo Zhang, and Ming-Hsuan Yang. **InstaInpaint: Instant 3D-Scene Inpainting with Masked Large Reconstruction Model**. In *NeurIPS*, 2025. [Project Page](#).
- [7] Xiaoyuan Wang, Yizhou Zhao, Botao Ye, Xiaojun Shan, **Weijie Lyu**, Lu Qi, Kelvin C.K. Chan, Yinxiao Li, and Ming-Hsuan Yang. **HoliGS: Holistic Gaussian Splatting for Embodied View Synthesis**. In *NeurIPS*, 2025.
- [8] Kuan-Chih Huang, **Weijie Lyu**, Ming-Hsuan Yang, and Yi-Hsuan Tsai. **PTT: Point-Trajectory Transformer for Efficient Temporal 3D Object Detection**. In *CVPR*, 2024. [Project Page](#).
- [9] Zhen Zhu*, **Weijie Lyu***, Yao Xiao, and Derek Hoiem. **Continual Learning in Open-vocabulary Classification with Complementary Memory Systems**. In *TMLR*, 2024. [Project Page](#).
- [10] Zhen Zhu, Yijun Li, **Weijie Lyu**, Krishna Kumar Singh, Zhixin Shu, Sören Pirk, and Derek Hoiem. **Consistent Multimodal Generation via A Unified GAN Framework**. In *WACV*, 2024.
- [11] Chejian Xu*, Wenhao Ding*, **Weijie Lyu**, Zuxin Liu, Shuai Wang, Yihan He, Hanjiang Hu, Ding Zhao, and Bo Li. **SafeBench: A Benchmarking Platform for Safety Evaluation of Autonomous Vehicles**. In *NeurIPS*, 2022. [Project Page](#).
- [12] Xianglong Tan, **Weijie Lyu**, and Andre Rosendo. **CircuitBot: Learning to Survive with Robotic Circuit Drawing**. In *PLOS ONE*, 2022.

Academic Service

Conference Reviewer: CVPR, ICCV, ECCV, ICLR, NeurIPS, ACM MM, ACM SIGGRAPH, etc.
Journal Reviewer: IEEE TPAMI, TMLR, ACM TOG, etc.